

Two Evaluation Traps in Constructed-Row Pipelines

Source-Item Leakage, Presentation-Order Shortcuts, and the Audit Protocol That Catches Them

Jan Kirin

github.com/VykosMolt | github.com/VykosMolt/Branching-Looped-Transformer

Paper 2 is the standalone methodology paper. Paper 1 contains the process-quality readout science and readout-control boundary; the erratum to arXiv:2604.09870 (v2) remains the correction of record.

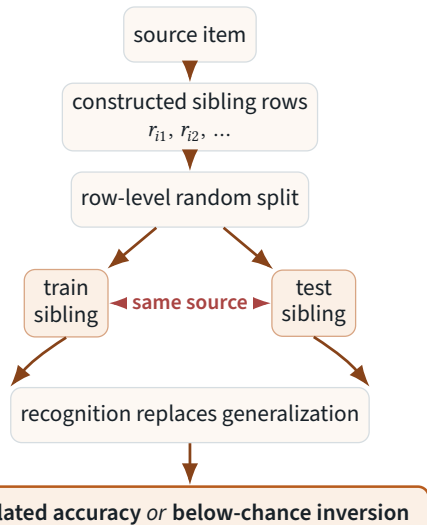
Abstract

Constructed-row evaluation pipelines can fail even when their split and scorer each look conventional. Auditing our own published and draft results revealed five distorted figures—four inflated and one deflated below chance, three in a published paper—caused by two distinct traps that are mutually blind. **Source-item leakage** occurs when multiple rows derived from one source item are randomly split by row: siblings cross the train/test boundary, and recognition replaces generalization. Depending on construction geometry, the error can inflate performance or invert it below chance. Four results failed this way, including a 21.75% “inverted polarity” result that corrects upward to 54.18%. Exact antisymmetry is not a safeguard against this leak: if a trained scorer fits one orientation and is exactly equivariant to the construction, its score on the transformed sibling is determined. Candidate-family transfer is a heuristic extension, not part of that exact proposition.

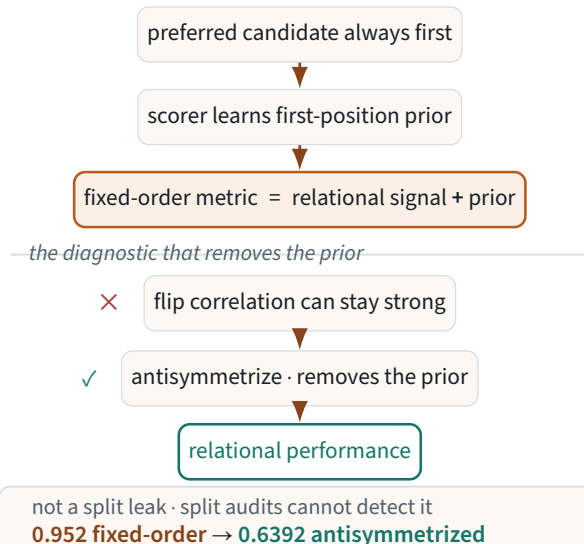
Presentation-order exploitation is not a split leak. A pairwise scorer trained, evaluated, or selected with the preferred candidate first can learn an order prior. In the audited case, fixed-order accuracy was 0.952, while strict antisymmetrized accuracy was 0.6392; normal/flipped score correlation remained -0.92 to -0.97 throughout training, so the usual flip diagnostic did not expose the shortcut. Synthetic experiments reproduce the traps’ qualitative signatures: leakage grows with model capacity and can appear above or below chance; an order-prior sweep preserves perfect flip correlation while separating fixed-order from relational accuracy. Near synthetic swap probability $q \approx 0.10$, fixed-order accuracy resembles the real evaluator, but the synthetic antisymmetrized value (approximately 0.56) is below the real 0.639. This is a qualitative artifact-state analogue, not an exact state match or mechanistic reconstruction; the real evaluator was trained at $q = 0.5$, and direct checkpoint-dynamics reconstruction remains future work.

We give a corrective protocol: split by source item and assert zero crossing; antisymmetrize every fixed-order pairwise evaluation; power the clean test; preserve checkpoints, raw predictions, seeds, and artifact hashes; and audit call sites rather than headline results. Reference tools implement the split and scorer checks. The correction of record remains the erratum; this paper treats the mechanisms and reusable audit method as a standalone methodology contribution.

TRAP 1 SOURCE-ITEM LEAKAGE



TRAP 2 PRESENTATION ORDER



Visual summary. Source-item leakage crosses splits; presentation-order exploitation does not. Both require separate audits.

Mechanisms and protocol at a glance

Auditing our own published and draft results, we found five distorted figures — four inflated, one deflated below chance; three of them in a published paper — produced by two distinct evaluation traps, each invisible to the other’s diagnostic. **Source-item leakage** arises when a dataset is built by constructing multiple rows from each source item ($\pm$ difference orientations, chosen/rejected singletons, candidate families, branches per task) and the constructed rows are split independently: the same source item lands on both sides of the train/test boundary, and held-out behavior becomes dependent on recognition of source items seen during training. Depending on construction geometry, that recognition can inflate accuracy or systematically invert predictions below chance. Four of the five figures were this, including one that presented not as a suspiciously good number but as an interestingly *bad* one — a 21.75% below-chance probe that a published paper interpreted as a finding (“inverted polarity”) and that corrects *upward* to 54.2%. We show that exact antisymmetry, a structural property we had relied on as a safeguard, *accelerates* this leak rather than preventing it: a probe that memorizes a training difference scores its exact negation correctly at evaluation precisely because it is antisymmetric — the leak transfers through the model class’s symmetry. **Presentation-order exploitation** is different in kind: a pairwise evaluator trained and evaluated with its candidates in a fixed order can score near-ceiling by learning to prefer the first-presented argument. Its data are properly split; no split check can see it; and the standard flip-test diagnostics do not catch it — the published case held a flip correlation of -0.92 to -0.97 at every training epoch while a third of its 95.2% accuracy was ordering prior (strict antisymmetrized accuracy: 63.9%).

We reconstruct both traps from fully synthetic data with known ground truth, reproducing the key signatures of the real cases: distortion that grows with probe capacity; the same defect presenting above *or* below chance depending on the construction’s geometry; and an order prior whose fixed-order accuracy is 0.95 while its antisymmetrized accuracy is 0.56, with flip correlation at -1.0 throughout. The synthetic order-prior experiment reproduces the real case’s key qualitative signature: near-ceiling fixed-order accuracy coexisting with substantially weaker antisymmetrized accuracy and perfect flip correlation. Near $q \approx 0.10$, it matches the real evaluator’s fixed-order accuracy, but not its exact antisymmetrized accuracy or its training dynamics. The real evaluator reached its audited state despite a full 50% swap protocol in training, because its checkpoint was selected by fixed-order accuracy while a transient prior was at its maximum — model selection on a fixed-order metric is itself an order-prior amplifier. We state the corrective protocol (source-item-disjoint splits with a zero-crossing integrity check; antisymmetrized evaluation for every fixed-order pairwise scorer; adequate power for the clean evaluation; auditing call sites rather than results), document a third failure class — non-reproducible provenance — with its own countermeasures, and release reference implementations of the two checks. Source-item leakage generalizes to evaluations built from augmentations, orientations, mutations, repeated measures, or candidate families of a shared source. Presentation-order exploitation generalizes to pairwise scorers trained, evaluated, or selected under a canonical candidate order.

1. Introduction

A machine-learning evaluation is trusted to the extent that its held-out set is genuinely held out and its metric genuinely measures the property claimed. This paper is about two ways those guarantees fail silently in a common and growing class of pipelines: evaluations whose rows are *constructed* — derived from source items by orientation, pairing, mutation, or expansion — rather than sampled directly. Preference learning constructs two rows per human-labeled pair. Contrastive and ranking evaluations construct candidate families per prompt. Program-repair and robustness benchmarks construct mutants per task. Search and branching evaluations construct many trajectories per problem. In every such pipeline there is a gap between the *row* (what the code splits) and the *source item* (what independence actually requires). In that gap lives the first trap. The second arises independently when pairwise candidates are trained, evaluated, or selected under a canonical presentation order.

We did not set out to study evaluation methodology. The traps were found the hard way, inside a research program probing the hidden states of a frozen looped transformer (Ouro-2.6B; Zhu et al., 2025), when a systematic audit of our own results found **five distorted figures — four inflated, one deflated below chance; three of them in a published paper** (Kirin, 2026a; corrected in the erratum to arXiv:2604.09870). The corrections are summarized in Figure 1. Four of the five were **source-item leakage**; the fifth was **presentation-order exploitation**, a mechanism different in kind: its data were split correctly, and the inflation lived in the presentation. Each trap is invisible to the other’s diagnostic — a split audit cannot see an ordering prior, and antisymmetrization cannot see a leak — which is how both survived, for months, in a project that was checking its work.

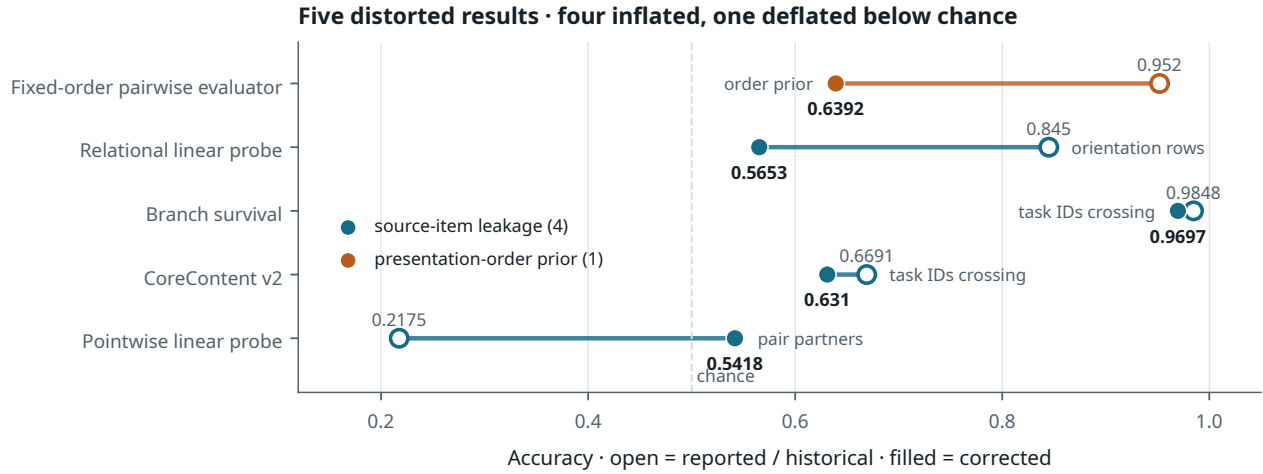


Figure 1. The five corrected results, colored by mechanism. Open markers are the reported values, filled markers the corrected ones. Note the pointwise probe: the same defect class that inflates four results deflates this one below chance — the artifact presented as a finding, not as an error.

This paper makes five contributions:

1. **A taxonomy of two traps with different mechanics, different diagnostics, and different fixes** (Sections 3–4), grounded in five fully documented real cases with pinned artifacts, including the published 95.2% → 63.9% and 84.5% → 56.5% corrections and a below-chance artifact that corrected upward.
2. **The safeguard-is-the-leak result** (Section 3.3): exact antisymmetry — enforced specifically to prevent evaluation gaming — is what makes orientation-row leakage maximally efficient. Structural constraints on the *scoring function* provide no protection against defects in the *dataset*, and can actively convert memorization into held-out success. We generalize: a construction map g leaks through any model class that commutes with g .
3. **Synthetic reconstructions of both traps** (Section 5) with known ground truth, reproducing the key signatures of the real cases from pure noise plus a weak planted signal: capacity-dependent inflation, above-chance and below-chance disguises of the same defect, and an order prior whose flip-test correlation is perfect while a third of its accuracy is artifact. The order-prior figure reproduces the real evaluator’s key qualitative signature and matches its fixed-order accuracy near $q \approx 0.10$, but not its exact antisymmetrized accuracy or its training dynamics; the mechanistic reconstruction is future work.
4. **A corrective protocol and reference tooling** (Sections 7–8): source-item-disjoint splitting with a zero-crossing integrity check that refuses to run otherwise; strict antisymmetrized evaluation for fixed-order pairwise scorers; power requirements for the clean evaluation; and the operational lesson that audits must enumerate *call sites*, not results.
5. **A third failure class — provenance** (Section 6): a published-adjacent claim whose central number had no surviving artifact and did not reproduce under controlled replication. Its countermeasures (pinned checkpoints and data revisions, preserved raw predictions, seeded and code-auditable statistics) are part of the same protocol.

Nothing here requires the host project’s subject matter. We use its cases because they are fully documented, because three of them were published and formally corrected, and because the project’s post-audit protocol has now produced a body of results that survived decontamination — evidence that the protocol is workable in practice, not merely prescriptive.

2. The host project, briefly

The cases come from a research program training small linear and near-linear probes (“taps”) on the frozen hidden states of a looped transformer to read properties of candidate continuations: human preference (from HH-RLHF pairs; Bai et al., 2022), content quality across five domains, branch survivability inside a live inference-time search scaffold, and generated-branch correctness. Two structural facts about this setting matter for everything that follows, and both are common far beyond it.

First, every dataset was built from constructed rows. A preference pair (chosen, rejected) became two orientation rows $+\Delta = h_{\text{chosen}} - h_{\text{rejected}}$ and $-\Delta$ for a relational probe, or two pointwise rows (one per candidate) for an independent probe. A content task became a family of candidate rows. A search problem became dozens of branch rows. In each case a generic row-splitting helper assigned constructed rows to train/eval independently at random.

Second, one evaluator was trained and evaluated in a fixed presentation order — the human-preferred candidate always first — because that is the natural way to write the code when the label is “the first one is better.”

Five headline figures emerged from this setting. All five were wrong:

Result	Construction	Reported	Corrected	Mechanism
Relational linear probe	2 orientation rows per pair	0.845	0.5653	source-item leakage (§3)
Pointwise linear probe	2 candidate rows per pair	0.2175 (below chance)	0.5418	source-item leakage (§3)
CoreContent content ranker	candidate families per task	0.6691	0.6310	source-item leakage (§3)
Branch-survival retention	branches per task	0.9848	0.9697	source-item leakage (§3)
Fixed-order pairwise evaluator	properly split pairs	0.952	0.6392 (antisymmetrized)	order exploitation (§4)

Every corrected value is verified against pinned repository artifacts: fixed dataset revisions, SHA-256-pinned model shards and evaluator checkpoints, fixed seeds, and 10,000-draw bootstrap intervals. Where a corrected value has a confidence interval it is reported in the sections below.

3. Trap 1: source-item leakage from constructed rows

3.1 The mechanism

Let source items $x_1, \dots, x_N$ be the units whose independence the evaluation is supposed to respect, and let each item generate $m \geq 2$ constructed rows $r_{i1}, \dots, r_{im} = g_1(x_i), \dots, g_m(x_i)$ through construction maps g_j . If the split operates on rows, then for a train fraction f the probability that a given evaluation row has at least one sibling in training is $1 - (1 - f)^{m-1}$ — at the standard $f = 0.8$ and just $m = 2$, that is already **80%** of evaluation rows. A model with enough capacity to memorize row-level structure can then make held-out behavior depend on *recognizing the source item*, without any generalizing knowledge. Depending on construction geometry, that dependence can inflate accuracy or systematically invert predictions below chance. The measured quantity silently changes from “does the probe generalize to unseen items?” to “how does the probe respond to items it recognizes?”

Three properties make this trap pernicious in practice:

- **The code looks correct.** A uniform random row split is the textbook operation. The defect is not in the splitting function but in the *unit* it is applied to, and the unit is invisible at the call site: a helper that correctly splits rows is a latent bug in every pipeline whose rows are constructed.
- **The magnitude of the distortion scales with model capacity** (demonstrated synthetically in §5.2), so the trap grows quietly as the evaluation’s probe or model is improved — progress on the model manufactures progress on the metric.
- **The artifact’s sign depends on the construction’s geometry** (§3.4, §5.3): the same defect can present as a suspiciously strong result *or* as a strikingly bad one. Only one of those triggers suspicion.

3.2 Four real cases

Case 1 — orientation rows (published as 84.5%). The relational preference probe scored pairwise differences with an exactly antisymmetric bias-free readout, $s(A, B) = w^\top(h_A - h_B)$. Its dataset contained both orientations of every pair, and the row split placed $+\Delta_i$ in training and $-\Delta_i$ in evaluation for most pairs. Corrected under a strict pair-disjoint split (40,000 source pairs, 32,000/8,000, zero pairs crossing, verified): **0.5653**. The mechanism of transfer is the subject of §3.3.

Case 2 — pair partners (published as 21.75%, below chance). The pointwise probe classified individual candidate rows as chosen/rejected. Both rows of each pair entered the row split independently, leaving 74% of evaluation rows with their pair partner in training. The probe learned per-pair directions whose polarity did not transfer to the held-out partner — producing a *below-chance* artifact that the published paper interpreted as “preference is not encoded as an absolute property of individual representations,” with inverted polarity as the pivotal diagnostic finding. Corrected pair-disjoint: **0.5418** (95% CI [0.5366, 0.5471]), significantly *above* chance. The finding was withdrawn; the corrected science is a modest, real relational-over-pointwise advantage (paired $\Delta +0.0234$, 95% CI [+0.0132, +0.0334]).

Case 3 — candidate families (draft-stage 0.6691). A content-quality ranker was evaluated on a stored split in which **195 task IDs** appeared on both sides of the train/held-out boundary — candidate families constructed per task, split per row. Corrected task-disjoint (zero crossing): **0.6310**, against a baseline of 0.5525. The effect survived decontamination

with a 3.8-point loss.

Case 4 — branches per task (draft-stage 0.9848). A branch-survival evaluation had 8 task IDs crossing the boundary, and 26 of its 48 evaluation tasks were training-side. Corrected: **0.9697**. Two points about this case carry general weight. It was found *after* the project had written a section about the error class and believed itself chastened — see §7 on call sites. And downstream figures that had rested on the contaminated split (terminal-selection magnitudes) could not be repaired, only withdrawn: decontamination left too few informative tasks to quantify the effect in either direction. Leakage does not merely inflate numbers; when corrected late, it can destroy the power of everything built on top.

3.3 The safeguard is the leak: antisymmetry accelerates it

Case 1 is the sharpest lesson of the four, because the probe carried a structural guarantee that we had explicitly relied on as evidence of trustworthiness. The scoring function is *exactly antisymmetric by construction*: $s(B, A) = -s(A, B)$ to machine precision, so the probe provably cannot prefer a candidate because of the side it was presented on. The reasoning error was to let that guarantee stand in for evaluation integrity in general. Exact antisymmetry constrains the scoring function; it says nothing about the *dataset*. And against orientation-row leakage it is worse than useless:

A probe that has memorized $w^\top \Delta_i > 0$ from a training row scores the evaluation row $-\Delta_i$ correctly **because** it is antisymmetric: $w^\top(-\Delta_i) = -w^\top \Delta_i < 0$. The memorized fact transfers to the held-out row with probability 1. The structural property that made the probe honest about presentation made it maximally efficient at exploiting the leaked split.

The general form is worth stating as a compact proposition:

Proposition (leakage through commutation). Let rows $r_{ij} = g_j(x_i)$ be constructed from source items x_i by construction maps that form a group action, with labels transforming under the same action as $y_{ij} = \rho_j(y_i)$. Suppose the *trained* scorer s satisfies the exact equivariance $s(g_j(x)) = \rho_j(s(x))$. Then, **conditional on s having fit one constructed row r_{ik} of a source item**, its score on every sibling r_{ij} related by the action is determined exactly. Row-level splitting therefore converts source-item memorization into correct held-out predictions on precisely those siblings. The premise does real work: it requires the trained scorer to interpolate the observed row and to satisfy the equivariance exactly. Orientation rows under a bias-free antisymmetric scorer are the case where both hold rigorously ($g(x) = -x$, $\rho = -1$, and $s(-x) = -s(x)$ by construction, so a probe that fits Δ_i scores $-\Delta_i$ correctly with certainty).

The extension to candidate families, mutants, and repeated branches is a **heuristic generalization**, not part of the proposition: there the construction is not an exact symmetry of the scorer, so transfer flows through shared features and is attenuated but rarely zero. We state it as a rule of thumb — the more nearly the model class commutes with the construction, the more completely a row-level split leaks — and verify both ends of it synthetically in §5.2.

The practical test falls out directly: *for each construction map g_j in the pipeline, ask whether the model class can express the relation between r_{ij} and the other rows of x_i* . If the trained scorer expresses the relation exactly and fits the observed row, transfer to the transformed sibling is exact. Approximate expression produces correspondingly attenuated leakage (though rarely zero — shared features leak through generic similarity too). Section 5.2 demonstrates both regimes: the antisymmetric probe inherits the leak completely, while a 1-nearest-neighbour memorizer — for which a negated row is geometrically *far* from its sibling — shows no inflation at all on the identical leaked split. The leak is a joint property of the construction and the model class, not of the data alone.

3.4 Phenomenology: how four leaks hid in an audited project

The four cases did not look alike from the inside, and the differences are the reason they survived.

The artifact does not announce itself as good news. Case 2's below-chance 21.75% *invited interpretation* rather than suspicion: below-chance results read as “signal with inverted polarity,” which is interesting, publishable, and wrong. In the synthetic reconstruction (§5.3) the same defect produces 0.91 or 0.18 depending only on how much of each row's variance is shared within the pair — the direction of the artifact carries no information about whether there is an artifact.

Suspicion is miscalibrated precisely because it is downstream of the leak. The figure that survived scrutiny longest (Case 4's 0.9848) was contaminated; the figure that looked most obviously broken (Case 2's 21.75%) was the leak announcing itself. Intuitions about which numbers “look fine” are formed by the same inflated landscape the leak created, and are therefore not evidence.

A heuristic that would have worked, offered with feeling: if a number is substantially better than every other number in the project, that is not a finding; it is a hypothesis about the evaluation. Both of this project's standout numbers (0.952, 0.845) were exactly that.

4. Trap 2: presentation-order exploitation

4.1 The mechanism

A pairwise evaluator is trained to score candidate pairs presented in a canonical order — the preferred candidate first — and evaluated the same way, with “score > 0” counted as correct. Nothing is leaked: the train and test pairs are disjoint, and every split check passes. But the presentation itself carries the label. Any capacity the model has for order-sensitive features — a bias term is already enough — can absorb a *first-argument prior*, and the fixed-order metric cannot distinguish that prior from genuine relational discrimination. What crosses the train/test boundary is not an item but the label, through the protocol.

Decompose any pairwise score into its order-independent and order-dependent parts,

$$s_{\text{anti}}(A, B) = \frac{1}{2} [s(A, B) - s(B, A)], \quad s_{\text{sym}}(A, B) = \frac{1}{2} [s(A, B) + s(B, A)].$$

Fixed-order accuracy scores $\text{sign}(s_{\text{anti}} + s_{\text{sym}})$; only s_{anti} carries relational information. **Strict antisymmetrized accuracy** — the fraction of pairs on which s_{anti} alone has the correct sign — is the metric that removes the prior, and it requires scoring both presentation orders.

4.2 The real case: 95.2% → 63.9%

The published evaluator (attention pooling, per-loop candidate differences, a trained difference-LayerNorm, a GRU, a nonlinear scorer) reported **95.2%** on the full 8,552-pair HH-RLHF test set — above end-to-end reward models trained on 3× the data, above the estimated annotator-agreement ceiling, and the headline of the paper. The audit, run on the pinned checkpoint over the full test set:

Measurement	Value
Fixed-order (canonical) accuracy	0.9479 (95% CI [0.9431, 0.9525]; reproduces the published 0.9519)
Swapped-order accuracy	0.1954
Strict antisymmetrized accuracy	0.6392 (95% CI [0.6291, 0.6493])
Symmetric component, mean	+1.2649
Symmetric / antisymmetric magnitude ratio	1.50×
Pairs scored “first argument” in both orders	75.25%
Normal/flipped score correlation	−0.9247
Strict sign-flip rate	0.2475

Roughly a third of the apparent accuracy was the ordering prior. The model is not degenerate — swapped scores are strongly anti-correlated with canonical ones, so content genuinely moves the score — but the learned first-position offset is on average 1.5× the content component and overwhelms the sign for three-quarters of pairs. Ablations on a 200-pair controlled slice localize the mechanism: replacing the learned attention pooling with masked mean pooling barely changes the numbers (pooling is not the cause), while *removing* the trained difference-LayerNorm makes order dominance stronger (canonical accuracy rises to 99–99.5%; the symmetric/antisymmetric ratio rises from ~1.6–2.0× to ~3.1–3.35×) — normalization had been partially *restraining* a prior the training objective was happy to grow.

Two aggravating details deserve emphasis. First — and it is the case’s sharpest lesson — the evaluator’s training already included a **full 50% swap protocol**, adopted specifically to prevent degenerate order solutions after an earlier version collapsed to a constant that scored 100% (Kirin, 2026a, §4.3). A 50% swap makes a *converged* order prior unlearnable in expectation, and the evaluator’s own published cross-epoch diagnostics confirm the offset was **transient** rather than converged: the mean normal-plus-flipped score rose from +1.04 (epoch 1) to +2.51 (epoch 2) and then decayed through +1.64, −0.22, to −0.37 by epoch 5 (Kirin, 2026a, Appendix C). The reported checkpoint was selected at epoch 2 by held-out **fixed-order** accuracy, which also peaks there (95.2%). Here is the trap in one observation: the two quantities peak at the *same epoch*. The prior’s magnitude is maximal exactly where fixed-order accuracy is maximal — which is not a coincidence but a consequence, since fixed-order accuracy is signal *plus* prior and is therefore maximized, in part, by whatever maximizes the prior. Kirin (2026a) attributes the epoch-2 accuracy peak to a genuine generalization effect (a learning-rate dead zone acting as early stopping), and that may well be a real component; the point is that a fixed-order selection metric *cannot separate* the two, and will bank the prior as if it were generalization every time they coincide. This adds a corollary to the protocol: **model selection on a fixed-order metric is itself an order-prior amplifier** — checkpoint selection should use antisymmetrized accuracy, and the audit must be applied to the *evaluated artifact*, never inferred from the training recipe (§4.3, §5.4). Second, the corrected value is not noise: 0.6392 remains the strongest clean preference readout in the host project, comfortably above the linear probe’s 0.5653. The trap did not manufacture a signal

from nothing — it doubled a modest real one, which is precisely why it was believable.

4.3 Why the standard diagnostics miss it

The published paper did not skip diagnostics. It ran flip tests at every epoch and reported them prominently: normal/flipped score correlation was stable at **−0.92 to −0.97 across all training epochs**, and the paper concluded that correlation was “the robust primary metric” for pairwise evaluator antisymmetry, with the residual offset interpreted as benign scorer bias that “does not affect preference discrimination accuracy under consistent argument ordering.” That interpretation was exactly backwards: under consistent ordering, the offset *is* the inflation.

The general lesson: **flip correlation certifies order sensitivity, not relational discrimination**. Correlation between $s(A, B)$ and $s(B, A)$ measures whether content moves the score antisymmetrically; it is blind to any constant or low-variance symmetric component, which is exactly where an order prior lives. In the synthetic reconstruction (§5.4) the flip correlation is **−1.0 identically at every point of the sweep** — perfect order sensitivity — while fixed-order accuracy runs up to 40 points above the true signal. Strict sign-flip rate fails in the same way (it tracks the prior’s magnitude, not the signal). The only diagnostic in this family that removes the prior rather than measuring around it is strict antisymmetrized accuracy, and it should be reported for **every** pairwise evaluator trained or evaluated in a fixed order — including ones whose training included swap protocols, which the real case shows are not sufficient.

5. Synthetic demonstrations

Real cases establish that the traps occur; synthetic reconstructions establish that the traps *suffice* — that nothing about the host project’s models or data is needed to produce the key audited end-state signatures. Each demonstration plants a weak global signal in high-dimensional noise, so ground truth is known exactly, and uses the same probe family and regularization as the real project (L2-regularized logistic readouts, $\lambda = 0.5\|w\|^2/n$). All results are means over 20 independent replicates with SDs; parameters and code are in the released demo suite.

5.1 Setup

Source pairs are differences $\Delta_i = \beta w^* + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, I_d)$: a weak true direction w^* whose oracle accuracy $\Phi(\beta)$ is known analytically. Constructed rows, splits, and evaluators then follow each trap’s recipe.

5.2 Orientation rows: inflation, and its dependence on the model class

Each pair contributes rows $(+\Delta_i, +1)$ and $(-\Delta_i, -1)$; the row-level split holds out 20% of rows; the pair-level split holds out 20% of pairs ($N = 200$ pairs, $d = 1024$, $\beta = 0.6$, oracle 0.726).

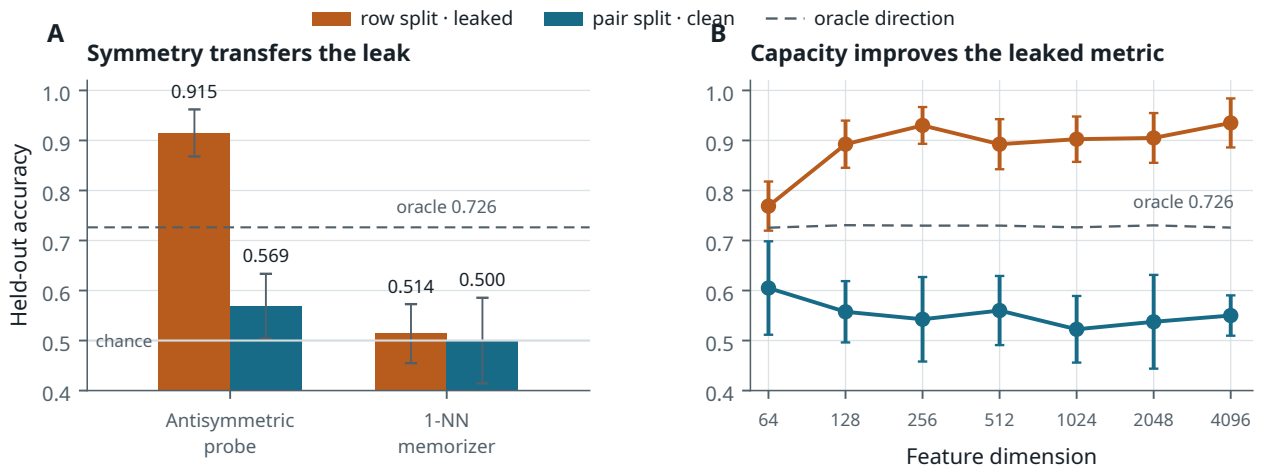


Figure 2. Left: the antisymmetric probe reads 0.915 on the leaked row split against its true pair-split generalization of 0.569 — while a 1-NN memorizer on the identical leaked split shows no inflation at all (0.514), because a negated row is geometrically far from its sibling. The leak transfers through the model class’s symmetry, not through the data alone. Right: inflation grows with probe capacity while clean generalization declines — improving the probe improves the broken evaluation.

Two signatures match the real Case 1. The inflation is large (+0.35 over the probe’s actual generalization) and **it increases with capacity** (right panel): as d grows past the interpolation threshold the leaked metric climbs toward

ceiling while the clean metric decays toward chance — a pipeline that “improves” its probe sees its leaked evaluation improve in lockstep. And the model-class contrast (left panel) isolates the §3.3 mechanism: the 1-nearest-neighbour memorizer, which cannot express the negation relation, gains nothing from the identical leak. Structural symmetry in the scorer is the transfer channel.

5.3 Pair partners: the same defect above and below chance

Pointwise rows are chosen $_i = \mu_i + s_i$ and rejected $_i = \mu_i - s_i$, where s_i carries a weak global signal ($\beta = 0.15$) and μ_i is a shared per-pair component of varying strength — the knob that controls the artifact’s disguise. Rows are split independently (79% of evaluation rows have their partner in training, closely matching the real case’s 74%).

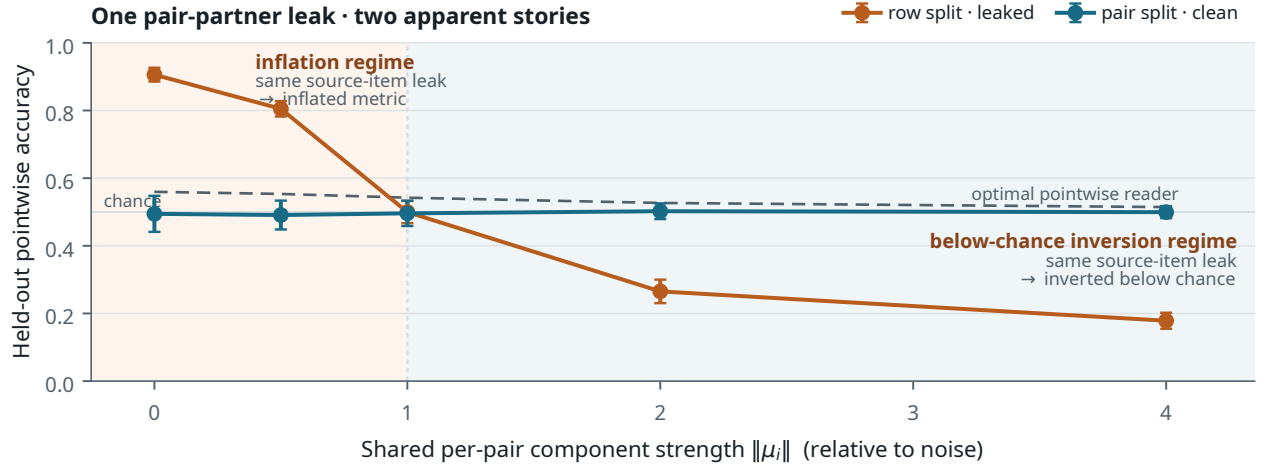


Figure 3. The same row-level defect presents as a discovery (0.91) when rows are near-antipodal within a pair, crosses chance, and presents as “inverted polarity” (0.18) as the shared per-pair component grows — the synthetic twin of the published 21.75%. The pair-disjoint split is flat and honest throughout; the optimal pointwise reader (dashed) shows how modest the true signal actually is.

The mechanism inverts cleanly. When μ_i dominates, the probe’s cheapest fit to a training row is the pair’s shared direction, labeled with *that row’s* label; the held-out partner has the same shared direction and the opposite label, and is therefore systematically misclassified — accuracy falls as far below chance as memorization is strong. When μ_i is negligible, the pair’s rows are near-antipodal and the leak works exactly like §5.2, inflating instead. The published below-chance finding needed no inverted polarity in the representations; it needed only this geometry, a row split, and enough capacity.

5.4 The order prior, under partial mitigation

A difference scorer with a bias path, $f(A, B) = w^\top(h_A - h_B) + b$, is trained with the preferred candidate first except for a fraction q of training examples presented swapped (with flipped target). The sweep varies the *training* swap probability q : at $q = 0$ the model is free to be a pure constant (the degenerate regime the real project caught with flip tests), and at $q = 0.5$ a converged order prior is unlearnable. ($N = 800$ pairs, $d = 512$, $\beta = 0.55$.)

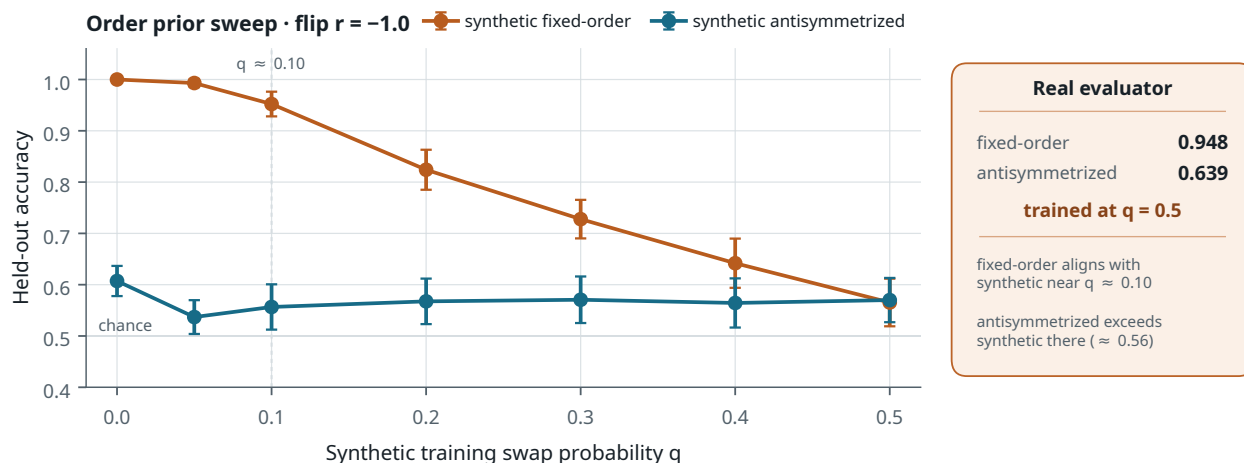


Figure 4. Fixed-order accuracy (orange) against strict antisymmetrized accuracy (blue) as the training swap probability q varies. The real evaluator’s fixed-order accuracy lies near the synthetic $q \approx 0.10$ point, while its antisymmetrized accuracy, 0.639, is higher than the synthetic mean of approximately 0.56. This is a qualitative end-state analogue, not an exact state match or a mechanistic reconstruction: the real evaluator was trained at $q = 0.5$, and reached its state by a different route — a transient order prior (largest at epoch 2) captured by fixed-order checkpoint selection (§4.2), not by partial training mitigation. Flip correlation is -1.00 at every point of the sweep, illustrating that order-sensitivity diagnostics cannot see the prior they coexist with.

The sweep makes three points compactly. The trap is not the degenerate constant — that is caught by existing flip tests — but the regime where real signal and a large prior coexist, which fixed-order accuracy cannot decompose. The correct metric is flat: strict antisymmetrized accuracy reads the same true signal at every q , because it never sees the symmetric component at all. And the *mechanism* by which the real $q = 0.5$ evaluator arrived at a large-prior state — a transient prior banked by fixed-order checkpoint selection — is represented here only by a qualitative end-state analogue, not an exact state match or its dynamics. A direct mechanistic reconstruction (train at $q = 0.5$, checkpoint over epochs, and show that selecting by fixed-order accuracy picks the transient prior maximum) is the clean confirming experiment, which we flag as future work rather than claim to have run.

6. A third failure class: provenance

One further correction in the host project was neither trap. A published-adjacent draft claimed a training-stage localization — that a fixed evaluator read the reasoning-trained model at $\sim 95\%$ but collapsed to 24% on the base model, implying reasoning fine-tuning *installs* the readable signal. Under controlled replication (pinned base revision, the original SHA-pinned evaluator checkpoint, identical tokenizer and inputs, both-order scoring), the base model reads $\sim 95\%$ canonical like every other backbone: **there is no collapse, and the historical 24% has no surviving artifact** — no command, no per-pair scores, no checkpoint identity. The claim was retracted rather than corrected: without the artifact, the cause (wrong checkpoint, wrong extraction locus, transcription error) cannot even be diagnosed.

The failure class is distinct: nothing was leaked and nothing was gamed — a number simply existed without the chain of custody that would let anyone, including its authors, re-derive it. Its countermeasures are the unglamorous half of the protocol: pin model revisions and checkpoint hashes; preserve raw per-example predictions alongside every reported aggregate (the host project’s later practice of shipping `*_rows.csv` with every audit made the antisymmetrization re-analysis possible at all); seed and preserve the statistics code, not just its output (an earlier confidence interval had to be recomputed from raw predictions because its bootstrap draws were not saved); and treat “no artifact” as “no claim” regardless of how well the number is remembered.

7. The corrective protocol

Consolidated from all three failure classes; every clean number in the host project and companion paper was produced under it.

1. **Split on source items, never constructed rows.** Identify the unit whose independence the evaluation claims (the pair, the task, the problem — not the row), and split there. Group-aware splitters exist in standard libraries (e.g. grouped k-fold); the failure mode is not the absence of tools but the invisibility of the unit at the call site.

2. **Enforce a zero-crossing integrity check.** Count source IDs appearing on both sides of the boundary and *refuse to run* at anything other than zero. This converts a silent defect into a loud one, and it is the difference between a convention and a guarantee.
3. **Antisymmetrize every fixed-order pairwise evaluation.** Score both presentation orders, decompose into symmetric and antisymmetric components, and report strict antisymmetrized accuracy alongside — or instead of — fixed-order accuracy. Report the symmetric/antisymmetric magnitude ratio as the size of the prior. Flip correlation and sign-flip rate are order-sensitivity diagnostics, not accuracy corrections (§4.3), and training-time swap protocols do not remove the need (§4.2, §5.4).
4. **Power the clean evaluation.** A source-disjoint split with $p \gg n$ returns chance whether or not signal exists; the host project’s first clean audit (800 pairs against 8,192 features) read 51% everywhere and was uninformative until rerun at 32,000 pairs. An underpowered clean evaluation can falsely convict a real result — decontamination and power are separate obligations.
5. **Audit call sites, not results.** A row-splitting helper applied to constructed data is a bug in a *function*, and functions have call sites. The host project fixed three results one at a time, wrote a section about the error, and then found the fourth leak under a result it had just promoted to load-bearing. Grep for the helper, enumerate every dataset it touches, and audit all of them before deciding which numbers to trust — because intuitions about which numbers look fine are downstream of the leak (§3.4).
6. **Pin provenance.** Model revisions and checkpoint hashes, dataset revisions, seeds, preserved raw per-example predictions, and code-auditable statistics (§6). A number without an artifact is a claim without evidence.
7. **Treat outliers as hypotheses about the evaluation.** If a figure is substantially better than every other figure in the project, audit it first.

Two meta-points. The checks are *mutually blind* — a project that audits splits and not presentation will find the leaks and keep the order prior, and vice versa; both real mechanisms survived months in a project running one check but not the other. And the protocol is cheap relative to what it protects: every item above is a constant-factor addition to an evaluation pipeline, while the cost of the alternative, in this project’s experience, was three published corrections, two withdrawn result families, and an erratum.

8. Reference tooling

Two small, dependency-light reference implementations accompany this paper (`tools/`):

- **`disjoint_split.py`** — `source_disjoint_split(source_ids, train_frac, seed)` returns row masks that are disjoint *by source item*, and `assert_zero_crossing(source_ids, train_mask, eval_mask)` raises on any source ID appearing on both sides. The assertion form matters: it is designed to be dropped into an existing pipeline directly before training, where it converts the silent defect into a hard failure.
- **`antisym_audit.py`** — `antisym_audit(score_fn, pairs)` scores both presentation orders and returns fixed-order accuracy, swapped-order accuracy, strict antisymmetrized accuracy, the strict sign-flip rate, normal/flipped correlation, and the symmetric/antisymmetric magnitude ratio — the full table of §4.2 for any pairwise scorer, in one call.

The synthetic demonstration suite (`demos/run_demos.py`) reproduces Figures 2–4 end-to-end from seeds and doubles as a test bed: a proposed pipeline defense can be checked against both traps by inserting it into the demos and verifying that the leaked and clean curves converge.

9. Related work

Leakage. Kaufman et al. (2012) give the classic formulation of leakage in data mining, including “leakage in training examples” through rows that share information with test rows; Kapoor & Narayanan (2023) survey leakage as a driver of the reproducibility crisis in ML-based science and provide a taxonomy in which our Trap 1 is a structured, construction-induced instance of train/test dependence. Elangovan et al. (2021) quantify train/test overlap effects in NLP evaluation. The closest prior instances to Trap 1 are in medical imaging, where splitting 2D slices rather than 3D subjects (“identity confounding”) is a recognized error. Yagis et al. (2021) report the same signature we reproduce synthetically — on randomly *labeled* brain-MRI data, slice-level splitting yields ~96% accuracy against the 50% that subject-level splitting correctly returns — and Bussola et al. (2021) document the tile-level analogue in digital pathology. We take this as external corroboration rather than as diminishing the point: a slice is a constructed row of a subject source item, so medical imaging has independently found the same trap in a different modality. What we add is (i) the algebraic characterization — the construction map’s symmetry determines when an exactly equivariant scorer that fits one sibling transfers that memorized prediction exactly to another, with antisymmetry as the sharp case (§3.3) — which the imaging literature’s

“identity confounding” describes only informally, and (ii) the observation that this *same* leak can present below chance, developed next.

Below-chance results. That a leaked evaluation can score *below* chance is not, on its own, novel: the neuroimaging MVPA literature documents systematic below-chance classification, attributed to small-sample cross-validation asymmetry (Jamalabadi et al., 2016) and to “anti-learnable” data geometry (Kowalczyk, 2007). Our contribution is to identify *paired-row source-item leakage* as a distinct cause, which we have not seen named as a below-chance mechanism in that literature: when a shared per-pair component dominates and the two rows carry opposite labels, the row split systematically inverts the held-out partner’s prediction, driving accuracy below chance as a direct function of leakage strength (§5.3). We claim novelty only cautiously — the below-chance literature is large — but the mechanism is distinct from the small-sample and anti-learning accounts, and the practical warning follows either way: a below-chance result should prompt a leakage audit, not only the small-sample corrections that literature recommends.

Grouped evaluation. Group-aware cross-validation is standard methodology (grouped k-fold and leave-one-group-out splitters ship in scikit-learn; Pedregosa et al., 2011), and fields with repeated measures per subject treat subject-level splitting as mandatory. Our contribution on this axis is not the splitter but the failure analysis — why competent practitioners running audits still miss the unit — and the zero-crossing assertion as an enforcement pattern rather than a convention.

Position and order bias. Order effects in pairwise judging are well documented for LLM-as-judge pipelines: Wang et al. (2023) show GPT-4-based comparative judgments flip with candidate order, and Zheng et al. (2023) measure position bias in MT-Bench-style evaluation, with randomized or both-order presentation as the standard mitigation. Trap 2 is the *trained* counterpart of that inference-time bias: the evaluator itself internalizes the canonical order of its training and evaluation protocol, and — as §4.2 shows — a training-time swap protocol can coexist with a large residual prior, so the audit must be applied to the trained artifact, not assumed from the training recipe. The decomposition into symmetric and antisymmetric score components, and strict antisymmetrized accuracy as the corrected metric, give the mitigation a quantitative form.

Benchmark contamination. The contemporary literature on benchmark contamination (e.g. Sainz et al., 2023; Zhou et al., 2023) concerns evaluation data appearing in pretraining corpora. Both traps here are disjoint from that problem — they arise entirely inside a single pipeline’s own construction and protocol — which is precisely why contamination-focused hygiene does not catch them.

10. Limitations

The case studies come from one research program and one modality (hidden-state probing of a language model), and the synthetic demonstrations use linear and near-linear probes; we have not characterized distortion magnitudes for deep evaluators trained end-to-end on constructed rows, where the §3.3 commutation analysis is only a heuristic. The taxonomy is not claimed to be exhaustive: two traps and a provenance class are what one audited project surfaced, and we see no reason to believe five is the number of ways an evaluation can silently distort — a project that audits for these three classes should still expect a fourth. Finally, the protocol’s costs are constant-factor but not zero (both-order scoring doubles pairwise evaluation compute; source-disjoint splits can reduce effective evaluation size and require the power analysis of §7.4), and we have not attempted to quantify the protocol’s adoption cost outside our own pipelines.

11. Conclusion

Five wrong numbers, two mechanisms, one project. The general conditions are ordinary: rows constructed from source items, and a pairwise protocol with a canonical order. Under those conditions a row-level split measures recognition rather than generalization — with structural safeguards like exact antisymmetry acting as accelerants, and with artifacts free to present as discoveries in either direction — while a fixed-order metric measures a prior plus signal with no way to tell the shares apart. Both traps pass the checks that competent projects actually run, and each is invisible to the other’s diagnostic.

The fixes are neither clever nor expensive: split on the source item and assert zero crossings; antisymmetrize every fixed-order pairwise evaluation; power the clean run; audit call sites; pin provenance. What they require is the belief that they are necessary *before* an audit proves it locally — which is why we have documented the real cases at full resolution, reconstructed them synthetically from known ground truth, and shipped the checks as code. The corrected story is less dramatic, and the surviving effects are real; that trade — less dramatic and real — is the entire argument.

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Appendix A — Synthetic demonstration parameters

Demo	Source pairs	Dim d	Signal β	Oracle acc.	Replicates	Figure
Orientation rows (bars)	200	1,024	0.60	0.726	20	Fig. 2 left
Inflation vs capacity	200	64–4,096	0.60	~0.72	10	Fig. 2 right
Pair partners	500	2,048	0.15	0.51–0.56 (pointwise)	20	Fig. 3
Order prior sweep	800	512	0.55	0.713	20	Fig. 4

All probes are L2-regularized logistic readouts matching the host project’s regularization ($0.5\|w\|^2/n$; scikit-learn $c=1.0$), except the $q = 0$ point of the order-prior sweep, which requires a one-class fit (gradient descent, unregularized bias) because a single-label dataset cannot be fit by standard two-class solvers. Splits hold out 20% (of rows or of source items, per condition). Seeds: base 20260716, per-replicate offsets recorded in `demos/run_demos.py`; results in `demos/demo_results.json`.

Appendix B — The five corrections, with provenance

Result	Reported	Corrected
Fixed-order pairwise evaluator	0.952	0.6392 [0.6291, 0.6493]
Relational linear probe	0.845	0.5653
Pointwise linear probe	0.2175	0.5418 [0.5366, 0.5471]
CoreContent ranker	0.6691	0.6310
Branch survival	0.9848	0.9697

Verification artifacts. Full 8,552-pair audit and SHA-pinned checkpoint: `full_hh_antisymmetry.json`; 40k-pair pair-disjoint rerun with zero crossings: `hh_thinking_40000_pair_disjoint_results.json`; pointwise pair-disjoint rerun: `hh_thinking_40000_pointwise_pair_disjoint_results.json`; task-disjoint CoreContent refit with zero crossing IDs: `ouro_corecontent_task_disjoint_results.json`; task-disjoint branch-survival rerun: `tier1_branch_survival_selection.json`.

The paired relational-over-pointwise comparison on identical held-out pairs is $\Delta = +0.0234$, 95% CI [+0.0132, +0.0334] (10,000-draw paired bootstrap, seed 20260711). All bootstraps use 10,000 draws.